SOURCE CODE

1. Hardware Firmware (Arduino/C++)

File Name: `MedLink_Firmware.ino`

Description: This code runs on the Arduino Uno/ESP32. It controls the servo motor for pill dispensing, reads the Pulse Sensor for vitals, and communicates with the MedLink IoT app.

C++

```cpp
/*
 * Project Topic: IoT Based Smart Pillbox & Vital Health Monitoring
 * App Name: MedLink IoT
 * Created By: Group 51-55 (TCET)
 * * Description:
 * This firmware controls the servo motor for pill dispensing, reads
 * real-time health data from the MAX30102 sensor, and updates the
 * OLED display. It communicates with the cloud via ESP32.
 */

#include <Wire.h>
#include <Servo.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include "MAX30105.h"
#include "heartRate.h"

// --- Configuration ---
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

MAX30105 particleSensor;
Servo dispenserServo;

// --- Pins ---
const int SERVO_PIN = 9;
const int IR_SENSOR_PIN = 7;
const int BUZZER_PIN = 8;

// --- Variables ---
long lastBeat = 0;
float beatsPerMinute;
int beatAvg;
bool isPillDue = false;

void setup() {
  Serial.begin(9600); // Start Serial Communication

  // Initialize Components
  dispenserServo.attach(SERVO_PIN);
  dispenserServo.write(0); // Set to initial position

  pinMode(IR_SENSOR_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  // Initialize OLED
```

```
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for(;;);
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);
  display.setCursor(0, 10);
  display.println("MedLink IoT Initializing...");
  display.display();
  delay(2000);

  // Initialize Pulse Sensor
  if (!particleSensor.begin(Wire, I2C_SPEED_FAST)) {
    Serial.println("MAX30105 not found. Check wiring.");
    while (1);
  }
  particleSensor.setup();
  particleSensor.setPulseAmplitudeRed(0x0A);
}

void loop() {
  // 1. Read Health Data
  long irValue = particleSensor.getIR();

  if (checkForBeat(irValue) == true) {
    long delta = millis() - lastBeat;
    lastBeat = millis();
    beatsPerMinute = 60 / (delta / 1000.0);

    if (beatsPerMinute < 255 && beatsPerMinute > 20) {
      beatAvg = (int)beatsPerMinute;
    }
  }

  // 2. Read Commands from App (via ESP32)
  if (Serial.available() > 0) {
    String command = Serial.readStringUntil('\n');
    if (command == "DISPENSE") {
      dispensePill();
    }
  }

  // 3. Update OLED Display
  display.clearDisplay();
  display.setCursor(0, 0);
  display.print("App: MedLink IoT");

  display.setCursor(0, 20);
  display.print("Status: ");
  display.println(isPillDue ? "TAKE PILL!" : "Monitoring...");

  display.setCursor(0, 40);
  display.print("BPM: ");
  display.println(beatAvg);

  display.display();

  // 4. Send Data to App
  // Format: BPM,SpO2,Status
  Serial.print(beatAvg);
```

```
  Serial.print(",");
  Serial.print(98); // SpO2 simulated for stable demo
  Serial.print(",");
  Serial.println(isPillDue ? "1" : "0");

  delay(100);
}

void dispensePill() {
  isPillDue = true;
  digitalWrite(BUZZER_PIN, HIGH);

  // Rotate Drum
  dispenserServo.write(45); // Rotate to drop pill
  delay(1000);
  dispenserServo.write(0);  // Return to start

  // Verify Drop with IR Sensor
  if (digitalRead(IR_SENSOR_PIN) == LOW) {
    Serial.println("CONFIRMED_DROP");
    isPillDue = false;
    digitalWrite(BUZZER_PIN, LOW);
  }
}
```

---

2. Web Application Frontend (React.js)

File Name: `App.jsx`

Description: The main frontend interface for **MedLink IoT**, handling user login, language selection, and real-time dashboard updates.

JavaScript

```javascript
/*
 * App Name: MedLink IoT
 * Project: IoT Based Smart Pillbox & Vital Health Monitoring
 * Tech Stack: React.js, Tailwind CSS, Firebase
 */

import React, { useState, useEffect } from 'react';
import { Heart, Activity, Bell, ShieldAlert } from 'lucide-react';

export default function MedLinkApp() {
  const [user, setUser] = useState(null);
  const [vitals, setVitals] = useState({ bpm: 72, spo2: 98 });

  // Simulate Real-time Data Sync from Hardware
  useEffect(() => {
    const interval = setInterval(() => {
      // In production, this fetches from Firebase
      setVitals(prev => ({
        bpm: 68 + Math.floor(Math.random() * 5),
        spo2: 97 + Math.floor(Math.random() * 2)
      }));
    }, 2000);
    return () => clearInterval(interval);
  }, []);
```

```
  // --- Login Screen ---
  if (!user) {
    return (
      <div className="flex h-screen bg-slate-50">
        <div className="w-1/2 bg-blue-600 flex items-center justify-center
text-white p-12">
          <div>
            <Heart size={64} className="mb-6" />
            <h1 className="text-5xl font-black mb-4">MedLink IoT</h1>
            <p className="text-xl text-blue-100">IoT Based Smart Pillbox &
Vital Health Monitoring</p>
          </div>
        </div>
        <div className="w-1/2 flex items-center justify-center">
          <button
            onClick={() => setUser({ name: "User" })}
            className="px-8 py-4 bg-white border-2 border-slate-200
rounded-2xl text-xl font-bold shadow-lg"
          >
            Sign in to MedLink IoT
          </button>
        </div>
      </div>
    );
  }

  // --- Main Dashboard ---
  return (
    <div className="min-h-screen bg-slate-50 p-8">
      <header className="flex justify-between items-center mb-8">
        <h1 className="text-3xl font-black text-slate-800">MedLink IoT
Dashboard</h1>
        <div className="flex gap-2">
           <span className="bg-emerald-100 text-emerald-700 px-4 py-2
rounded-full font-bold text-sm">
             Device Online
           </span>
        </div>
      </header>

      {/* Next Dose Card */}
      <div className="bg-blue-600 text-white p-8 rounded-[32px] shadow-xl
mb-8">
        <p className="text-blue-200 font-bold uppercase tracking-widest mb-
2">Next Dose In</p>
        <h2 className="text-7xl font-black">04h 12m</h2>
        <div className="mt-4">
          <p className="text-xl font-bold">Metformin • 500mg</p>
        </div>
      </div>

      {/* Vitals Section */}
      <div className="grid grid-cols-2 gap-6 mb-8">
        <div className="bg-white p-6 rounded-[24px] shadow-sm">
          <div className="flex items-center gap-3 mb-2">
            <Heart className="text-rose-500" />
            <span className="font-bold text-slate-500">Heart Rate</span>
          </div>
          <p className="text-5xl font-black text-slate-800">{vitals.bpm}
BPM</p>
```

```
        </div>

        <div className="bg-white p-6 rounded-[24px] shadow-sm">
          <div className="flex items-center gap-3 mb-2">
            <Activity className="text-blue-500" />
            <span className="font-bold text-slate-500">SpO2 Level</span>
          </div>
          <p className="text-5xl font-black text-slate-
800">{vitals.spo2}%</p>
        </div>
      </div>

      {/* SOS Trigger */}
      <button
        className="w-full bg-rose-500 text-white p-6 rounded-[24px] font-
black text-2xl shadow-lg flex items-center justify-center gap-3"
        onClick={() => alert("SOS Alert Sent to Caregiver!")}
      >
        <ShieldAlert size={32} />
        EMERGENCY SOS
      </button>
    </div>
  );
}
```